

CHAPTER I

INTRODUCTION

Background of the Study

Agriculture has long been the backbone of rural livelihoods in the Philippines, particularly in the province of Pampanga, where it serves not only as a primary economic activity but also as a cultural and social tradition. For generations, farmers have relied on inherited knowledge, intuition, and seasonal patterns to guide critical decisions such as planting, irrigating, and harvesting. However, the accelerating impacts of climate change have begun to disrupt these long-established practices. Increasingly erratic weather patterns such as delayed rainfall, prolonged dry seasons, and intensified typhoons that have made farming more unpredictable and vulnerable to failure. According to the Department of Agriculture–Philippine Rice Research Institute (2025), over ₱1 billion worth of crop losses in Central Luzon were attributed to sudden climate-induced weather disruptions.

During these natural disasters, farmers' needs are immediate and twofold: ex-ante (preventative) and ex-post (recovery). According to a study by the Philippine Institute for Development Studies (PIDS), the most preferred mitigation tools include localized climate information and accessible crop insurance (PIDS, 2009). Post-disaster, farmers require immediate access to aid such as seed and fertilizer subsidies, rehabilitation funds, and clear information on how to access government support programs (ERIA, 2013). AgriKlima is strategically designed to address the critical information gap in both phases. It serves as an ex-ante tool by providing advanced, localized weather warnings from OpenWeather and Open-Meteo, and as an ex-post information hub by using its 'News' module to instantly disseminate LGU announcements on aid distribution, available subsidies, and crop recovery programs.

Despite the availability of global advancements in agricultural technology, many Filipino farmers particularly those in rural and low-resource areas are still lacking access to timely weather data, decision-making tools, and digital farming support. As a result, key

farming activities are often mistimed, leading to reduced crop yields, wastage of resources, and income instability. Lapidus et al. (2024) emphasize the importance of real-time, interactive climate-based tools in supporting agricultural adaptation, particularly among farmers in resource-constrained communities

This reliance on intuition in the face of erratic weather, while traditional, often leads to mistimed planting, inefficient resource application, and significant post-harvest losses, all of which directly suppress farm productivity. The lack of accessible, real-time data means critical decisions are often reactive rather than proactive. As noted by the Philippine Institute for Development Studies (PIDS), digital agriculture offers significant potential to enhance productivity by facilitating data-driven farming practices and modernizing the food value chain (Briones et al., 2023). AgriKlima is designed to be this digital tool, bridging the information gap to boost productivity at the farm level.

Ultimately, AgriKlima is more than a digital platform. It is a strategic intervention aimed at empowering rural farmers through information. By facilitating data-driven agricultural practices, the system supports the broader goals of food security, climate adaptation, and inclusive technological access in Philippine agriculture.

Project Context

The unpredictable impacts of climate change are increasingly posing problems for farming communities in rural regions of the Philippines. Income streams and productivity levels are becoming increasingly difficult to secure due to traditional farming practices, which require more reliance on historical weather patterns and experiential knowledge. Addressing the issues posed by changing climates, AgriKlima project gives local farmers tools by offering accessible and reliable digital climate data. This system is aimed at equipping farmers with reliable information that aims the local farmers to make informed decisions, adapt techniques, and become more resilient to climate threats. Besides providing a technological remedy, AgriKlima advocates a shift towards sustainable agriculture that embraces proactive strategies, thus empowering farmers to protect their livelihoods while enhancing food security even with fluctuating climate conditions.

Agriculture continues to be a critical economic activity in District 3 of Pampanga. Yet, farmers are struggling with financial instability due to weather-related losses and loss of crops. AgriKlima aims to reduce economic risk by assisting farmers in making smarter decisions based on accurate data that help avert unnecessary resource (seeds, water, and fertilizer) wastage. This system provides data to support crop yield efficiency, which in turn can contribute to income stability and sustained agricultural output, fostering economic growth at the local level.

Due to climate change, farming has become more difficult because of prolonged dry spells, excessive rainfall, and stronger typhoons. These issues erode the soil as well as decrease biodiversity in farmland ecosystems. AgriKlima is committed to environmental conservation through advocating for smart farming techniques such as weather-resilient crop selection and optimal planting schedules. It also helps with advanced warning concerning vermin and plant ailments, which when controlled minimizes chances of chemical overuse and over-farming.

AgriKlima ensures long-term success for farmers through providing farmers with real-time weather information, previous climate patterns, and specialized knowledge in the day-to-day decisions. It encourages clever and adaptable farming practices that are effective even when the weather varies. By providing practical, localized assistance, AgriKlima enables farmers to care for their land, guard their crops, and maintain a healthy environment. Ensuring that agriculture remains resilient for generations to come.

By ensuring that all farmers, especially those in rural or impoverished locations, have access to useful tools and information, the crucial ethical consideration advances justice. Even people who are not inclined to technology may easily utilize it, and it doesn't promote any biased or commercial information. The method emphasizes the traditional significance of farming in rural Filipino communities and supports farmers' autonomy. Additionally, it safeguards farmers' private information, ensuring that it is never disclosed without consent.

Even without official training or assistance, AgriKlima helps bridge the gap between rural farmers and contemporary tools by providing simple access to crucial meteorological and farming information. Additionally, AgriKlima promotes cooperation among farmers, agricultural laborers, and neighborhood organizations, assisting communities in remaining resilient and cohesive as these groups jointly confront the consequences of climate change.

AgriKlima supports the government initiatives for agricultural modernization and climate adaptation. It can help local government entities and the Department of Agriculture in their efforts to reduce the risk of disasters and ensure food security. The program offers information that could help policymakers spot agricultural trends and better plan responses. AgriKlima can potentially be used as a prototype for upcoming government-supported digital farming tools and platforms by showcasing a workable and scalable solution.

AgriKlima aims to connect climate information with everyday farming practices by offering practical, localized guidance for farmers in District 3 of Pampanga. Its purpose is to enhance agricultural productivity and profitability by providing reliable weather updates, climate-based crop recommendations, and pest/disease alerts. By transforming data into simple, actionable insights, AgriKlima empowers farmers to make informed, data-based decisions that optimize resource use, reduce crop losses, and promote sustainable, productive farming.

Statement of the Objectives

To develop and evaluate AgriKlima, a localized farming support system designed to strengthen climate resilience, improve agricultural decision-making, and provide a centralized source of localized farming information for farmers in District 3 of Pampanga.

Specifically, this study aims to:

1. To identify the major climate and farming challenges affecting farmers in the Third Congressional District of Pampanga through a needs assessment survey and interviews.

2. To design and implement a user-friendly, localized digital platform (AgriKlima) that provides real-time weather forecasts, climate-based crop recommendations, and pest information.
3. To evaluate the technical quality and usability of the AgriKlima system through expert review by IT professionals

Significance of the Study

AgriKlima addresses the critical need for timely, accurate, and localized climate information in agriculture, particularly farmers in District 3 of Pampanga. In the face of unpredictable weather patterns, pest outbreaks, and declining crop yields, the system serves as a digital advisory platform that promotes data-driven decision-making and climate resilience. By integrating weather forecasts, crop recommendations, and pest alerts into one accessible tool, AgriKlima contributes to sustainable agriculture, informed decision-making, and better risk management.

Local Farmers. Farmers in District 3 of Pampanga gain access to real-time weather updates, pest alerts, and crop recommendations. This directly improves their productivity and profitability by empowering them with data to reduce crop losses from pests and extreme weather, and optimize the use of expensive inputs like fertilizer and water. The system helps them plan and adapt more effectively, leading to improved productivity, more stable incomes, and better management of farm operations.

Local Government Units (LGUs). AgriKlima can serve as a decision-support tool for LGU officers and agricultural departments, providing the LGU with localized climate and farming data that can be used for more informed planning and resource allocation. The system can also aid in early warning systems and proactive measures during extreme weather events, helping LGUs minimize agricultural damage and support their constituents more effectively.

Agricultural Extension Workers. Extension officers will be equipped with an evidence-based advisory tool that supports more accurate, timely, and standardized guidance for farmers in the field.

Future Researchers and Information System Developers. AgriKlima offers a scalable framework for research and development in climate-smart agriculture, enabling studies on system effectiveness and innovation in AI and user experience design.

Students and Academicians. The project serves as a real-world case study for students in agriculture, environmental science, and information systems, promoting interdisciplinary learning and project-based applications.

Private Sector in Agriculture and Agri-Tech Startups. Companies and startups can explore AgriKlima as a proof of concept for partnerships, innovation, or service integration in digital farming, crop insurance, and agricultural input distribution.

Scope and Delimitation

This study focuses on the design, development, and evaluation of AgriKlima, a localized farming aid system intended to support farmers in District 3, Pampanga, specifically the municipalities of San Fernando City, Arayat, Bacolor, Mexico, and Santa Ana. The system provides localized, real-time agricultural recommendations by integrating weather forecasts from OpenWeather, and Open-Meteo APIs, pest and disease alerts, and crop cultivation suggestions based on current and historical environmental data. AgriKlima is designed to be accessible via mobile devices, tablets, and desktop computers through standard web browsers, supporting both English and Filipino languages to accommodate users with varying levels of digital literacy and educational attainment identified through the needs assessment phase.

The scope of the study centers on assessing AgriKlima's suitability as a tool to support climate-adaptive farming decisions. The study's premise is that for a system to help farmers improve productivity and reduce losses, it must first be technically sound and usable. Therefore, this study evaluates the foundational quality, technical soundness, and usability of the system as judged by IT professionals. The system primarily caters to local farmers cultivating rice, corn, and vegetable crops which are the most commonly grown crops in the region based on the needs assessment survey showing 89% cultivating rice, 72% growing corn, and 18% planting vegetables. System

recommendations are optimized for these agricultural profiles and localized climate conditions specific to District 3 municipalities.

The study is explicitly delimited to farmers residing in District 3 of Pampanga, specifically San Fernando City, Arayat, Bacolor, Mexico, and Santa Ana, and does not extend to other districts within Pampanga province or to other agricultural regions in Central Luzon or the Philippines.

The AgriKlima system's weather forecasts and climate recommendations are dependent on the accuracy, timeliness, and availability of data from external sources such as the OpenWeather API and Open-Meteo API. The system functions as an information aggregation and presentation platform rather than generating original meteorological data, and limitations in these sources' forecasting accuracy, geographic coverage granularity, update frequency, or service availability (including API downtime or rate limiting) may directly affect the quality, reliability, and usefulness of system recommendations provided to farmers. The system cannot compensate for inaccuracies in the underlying source data.

Furthermore, to clarify the scope of Objective 2, the term 'implement' refers to the successful development and technical validation of the AgriKlima system as a functional prototype. The study is methodologically delimited to this development and the subsequent technical and usability evaluation of the AgriKlima system by expert respondents (IT professionals).

This study does not extend to the deployment or implementation of the system in a live field setting with the primary end-users (farmers and MAOs). It does not include a longitudinal field study to measure the quantitative impact on actual crop yields or a direct financial analysis of reduced agricultural losses. Therefore, while the system is designed to enable the enhancement of climate decisions, improvement of productivity, and reduction of losses, the scope of this research is to validate the system's quality and readiness for a future pilot deployment, not the direct measurement of these real-world agricultural outcomes.